

Peidong Wang

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Education

Doctor of Philosophy (Ph.D.)

Cambridge, MA

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Sep. 2019 - May. 2025

- Major: Climate Science
- Advisor: Susan Solomon
- Thesis: Understanding Drivers of Stratospheric Ozone Change and Fingerprinting its Recovery

Bachelor of Science (B.S.)

Madison, WI

UNIVERSITY OF WISCONSIN-MADISON

Sep. 2015 - May. 2019

- Major: Atmospheric & Oceanic Sciences
- Major: Applied Mathematics
- Minor (certificate): Computer Science
- Advisor: Tracey Holloway

Professional Experience

Stanford Science Fellow

Palo Alto, CA

STANFORD UNIVERSITY

Aug. 2025 - Present

Host: Noah Diffenbaugh

Postdoctoral Researcher

Cambridge, MA

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Jun. 2025 - Aug. 2025

Host: Susan Solomon

Graduate Research Assistant

Cambridge, MA

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Sep. 2019 - May. 2025

Advisor: Susan Solomon

Summer Student Fellow

Woods Hole, MA

WOODS HOLE OCEANOGRAPHIC INSTITUTION

May. 2018 - Aug. 2018

Advisor: Caroline Ummenhofer

Undergraduate Research Assistant

Madison, WI

UNIVERSITY OF WISCONSIN-MADISON

Sep. 2016 - May. 2019

Advisor: Tracey Holloway

Publications

In Prep / Submitted / Under Review

Peidong Wang, Susan Solomon, Clara Deser, David W. J. Thompson, Noah S. Diffenbaugh. **Antarctic Ozone Loss Shapes Surface Cooling Pattern and Climate Sensitivity.** [Preprint Link](#)

2025

Jian Guan, Benjamin D. Santer, Peidong Wang, Qiang Fu, Rolando R. Garcia, Yaowei Li, Kane Stone, Douglas Kinison, Jun Zhang, Gabriel Chiodo, Jean-Francois Lamarque, Susan Solomon. **Human Influence on the Ozone Layer Detectable by the 1960s.** *Proceedings of the National Academy of Sciences*, 123, e2608286123, 2026. [Paper Link](#), [News](#), [Nature Research Highlight](#)

Peidong Wang, Susan Solomon, Jeffery R. Scott, Shari A. Yvon-Lewis, Paul O. Wennberg, Ray F. Weiss, Matt Rigby, Minde An. **Ocean Outgassing of Methyl Chloroform as an Underestimated Source of Emission.** *Geophysical Research Letters*, 52, e2025GL118618, 2025. [Paper Link](#)

Aodhan Sweeney, Qiang Fu, Susan Solomon, Stephen Po-Chedley, William J. Randel, Andrea Steiner, Pu Lin, Thomas Birner, Sean Davis, Peidong Wang. **Recent Warming of the Southern Hemisphere Subtropical Lower Stratosphere and Antarctic Ozone Healing.** *AGU Advances*, 6, e2025AV001737, 2025. [Paper Link](#)

Peidong Wang, Susan Solomon, Benjamin D. Santer, Douglas E. Kinnison, Qiang Fu, Kane A. Stone, Jun Zhang, Gloria L. Manney, Luis F. Millán. **Fingerprinting the Recovery of Antarctic Ozone.** *Nature*, 639(8055), 2025. [Paper Link](#), [News](#)

2024

Peidong Wang, Susan Solomon. **Contrasting Chlorine Chemistry on Volcanic and Wildfire Aerosols in the Southern Mid-Latitude Lower Stratosphere.** *Geophysical Research Letters*, 51(18), 2024. [Paper Link](#)

Jun Zhang, Peidong Wang, Douglas Kinnison, Susan Solomon, Jian Guan, Yunqian Zhu. **Stratospheric chlorine processing after the unprecedented Hunga Tonga eruption.** *Geophysical Research Letters*, 51(17), 2024. [Paper Link](#)

2023

Glenn Liu, Peidong Wang, Young-Oh Kwon. **Physical Insights from the Multidecadal Prediction of North Atlantic Sea Surface Temperature Variability Using Explainable Neural Networks.** *Geophysical Research Letters*, 50(24), 2023. [Paper Link](#)

Peidong Wang, Susan Solomon, Megan Lickley, Jeffery Scott, Ray Weiss, Ronald Prinn. **On the influence of hydroxyl radical changes and ocean sinks on estimated HCFC and HFC emissions and banks.** *Geophysical Research Letters*, 50(18), 2023. [Paper Link](#)

Peidong Wang, Susan Solomon, Kane Stone. **Stratospheric chlorine processing after the 2020 Australian wildfires derived from satellite data.** *Proceedings of the National Academy of Sciences*, 120(11), 2023. [Paper Link](#)

Susan Solomon, Kane Stone, Pengfei Yu, Daniel Murphy, Doug Kinnison, AR Ravishankara, Peidong Wang. **Chemical impacts of wildfire smoke on stratospheric chlorine and ozone depletion.** *Nature*, 615(7951), 2023. [Paper Link](#), [News](#)

2022

Peidong Wang, Janni Yuval, Paul A. O’Gorman. **Non-local parameterization of atmospheric subgrid processes with neural networks.** *Journal of Advances in Modeling Earth Systems*, 14(10), 2022. [Paper Link](#)

Peidong Wang, Tracey Holloway, Matilyn Bindl, Monica Harkey, Isabelle De Smedt. **Ambient Formaldehyde over the United States from Ground-Based (AQS) and Satellite (OMI) Observations.** *Remote Sensing*, 14(9), 2022. [Paper Link](#)

2021

Glenn Liu, Peidong Wang, Matthew Beveridge, Young-Oh Kwon, Iddo Drori. **Predicting Atlantic Multidecadal Variability.** *NeurIPS 2021 Workshop on Tackling Climate Change with Machine Learning*, 2021. [Awarded with Best Paper: Pathway to Impact]. [Paper Link](#)

Megan Lickley, Susan Solomon, Doug Kinnison, Paul Krummel, Jens Mühle, Simon O’Doherty, Ronald Prinn, Matthew Rigby, Kane A Stone, Peidong Wang, Ray Weiss, Dickon Young. **Quantifying the Imprints of Stratospheric Contributions to Interhemispheric Differences in Tropospheric CFC-11, CFC-12, and N₂O Abundances.** *Geophysical Research Letters*, 48(15), 2021. [Paper Link](#)

Peidong Wang, Jeffery R. Scott, Susan Solomon, John Marshall, Andrew R. Babbitt, Megan Lickley, David W. J. Thompson, Timothy DeVries, Qing Liang, Ronald G. Prinn. **On the Effects of the Ocean on Atmospheric CFC-11 Lifetimes And Emissions.** *Proceedings of the National Academy of Sciences*, 118(12), 2021. [Paper Link](#), [News](#)

Thesis

Peidong Wang. **Understanding Drivers of Stratospheric Ozone Change and Fingerprinting its Recovery.** *Massachusetts Institute of Technology*, 2025. [Thesis Link](#)

Honors & Awards

- 2025–2028 **Stanford Science Fellow**, Stanford University
- 2025 **Carl-Gustaf Rossby Award for the Best PhD Thesis**, PAOC, MIT
- 2024–2025 **Jule Charney Fellowship**, PAOC, MIT
- 2023 **Chinese Government Award for Outstanding Self-Financed Students Abroad**, with the extraordinary prize for the top 20 awardees
- 2022 **Early Career Scientist Best Oral Presentation**, 7th SPARC General Assembly
- 2021–2022 **Norman C. Rasmussen Fellowship**, EAPS, MIT
- 2020–2021 **John H. Carlson Fellowship**, EAPS, MIT
- 2019 **Jule Charney Prize**, PAOC, MIT

- 2019–2020 **MIT Presidential Fellowship**, MIT
- 2019 **Herfurth-Kubly Awards for Comprehensive Undergraduate Excellence**, UW–Madison
- 2019 **University Book Store Excellence Award**, UW–Madison
- 2018 **Lyle Horn Scholarship Award**, AOS, UW–Madison
- 2018 **Summer Student Fellowship**, Woods Hole Oceanographic Institution
- 2017–2018 **Wisconsin Hildale Undergraduate/Faculty Research Fellowship**, UW–Madison

Teaching

12.003 Introduction to Atmosphere, Ocean, and Climate Dynamics

Fall 2021

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Teaching Assistant

Department of Earth, Atmospheric and Planetary Sciences

CS200 Programming I

Fall 2017

UNIVERSITY OF WISCONSIN–MADISON

Teaching Assistant

Department of Computer Sciences

CS302 Introduction to Programming

Spring 2017

UNIVERSITY OF WISCONSIN–MADISON

Teaching Assistant

Department of Computer Sciences

Selected Presentations

Invited talks

A Two-Way ‘Conversation’ Between the Stratosphere and the Ocean. Caltech, June 2026.

Antarctic ozone loss shapes surface cooling pattern and climate sensitivity. ECS & cloud feedback virtual symposia (online), May 2026.

From Stratospheric Ozone to the Ocean: Signals Behind the Patterns. CU Boulder, April 2026.

Fingerprinting the Recovery of Antarctic Ozone. Peking University, Beijing (online), Mar 2025.

Contrasting stratospheric chlorine processes on volcanic and wildfire aerosols. Jinan University, Guangzhou, Jan 2024.

Conference presentations

JpGU-AGU Joint Meeting (Oral), Chiba, Japan, May 2026.

AGU Fall Meeting (Oral), New Orleans, Louisiana, U.S., Dec 2025.

CESM Working Group Meeting (Oral), Boulder, Colorado, U.S., Feb 2025.

Quadrennial Ozone Symposium (Poster), Boulder, Colorado, U.S., Jul 2024.

EGU General Assembly (Oral), Vienna, Austria, Apr 2024.

AGAGE 68th Meeting (Oral), Boston, Massachusetts, Oct 2023.

VollImpact Summer School (Poster), Greifswald, Germany, Sep 2023.

SPARC 7th General Assembly (Oral), Boulder, Colorado, Oct 2022.

NeurIPS 2021 Workshop on Tackling Climate Change with Machine Learning (Spotlight talk), virtual, Dec 2021.

Quadrennial Ozone Symposium (Poster), online, Oct 2021.

AGU Fall Meeting (Poster), Washington D.C., Dec 2018.

Symposium in Honor of Isaac Held (Poster), Princeton University, NJ, Oct 2018.

HAQAST 3rd Meeting, Columbia University, NY, Nov 2017.

Academic Service

Committee Member PAOC Colloquium Committee (2021-2025)

Proposal Reviewer NASA ROSES panelist (2024)

Journal Reviewer *Advances in Climate Change Research · Communications Earth & Environment · Earth's Future · Environmental Science & Technology · Geophysical Research Letters · Geoscience Letters · Global and Planetary Change · Nature Communications · Remote Sensing of Environment · Science Advances*